

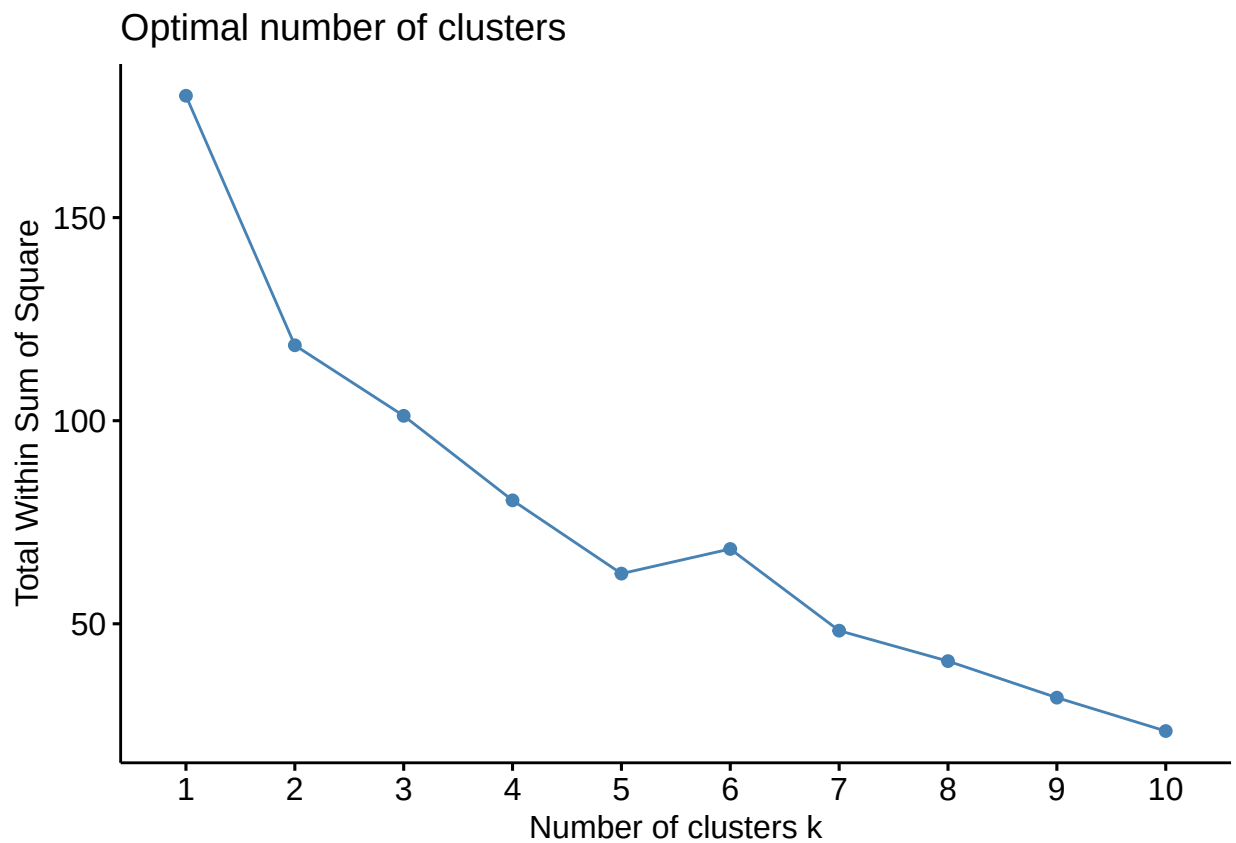
Assignment 4

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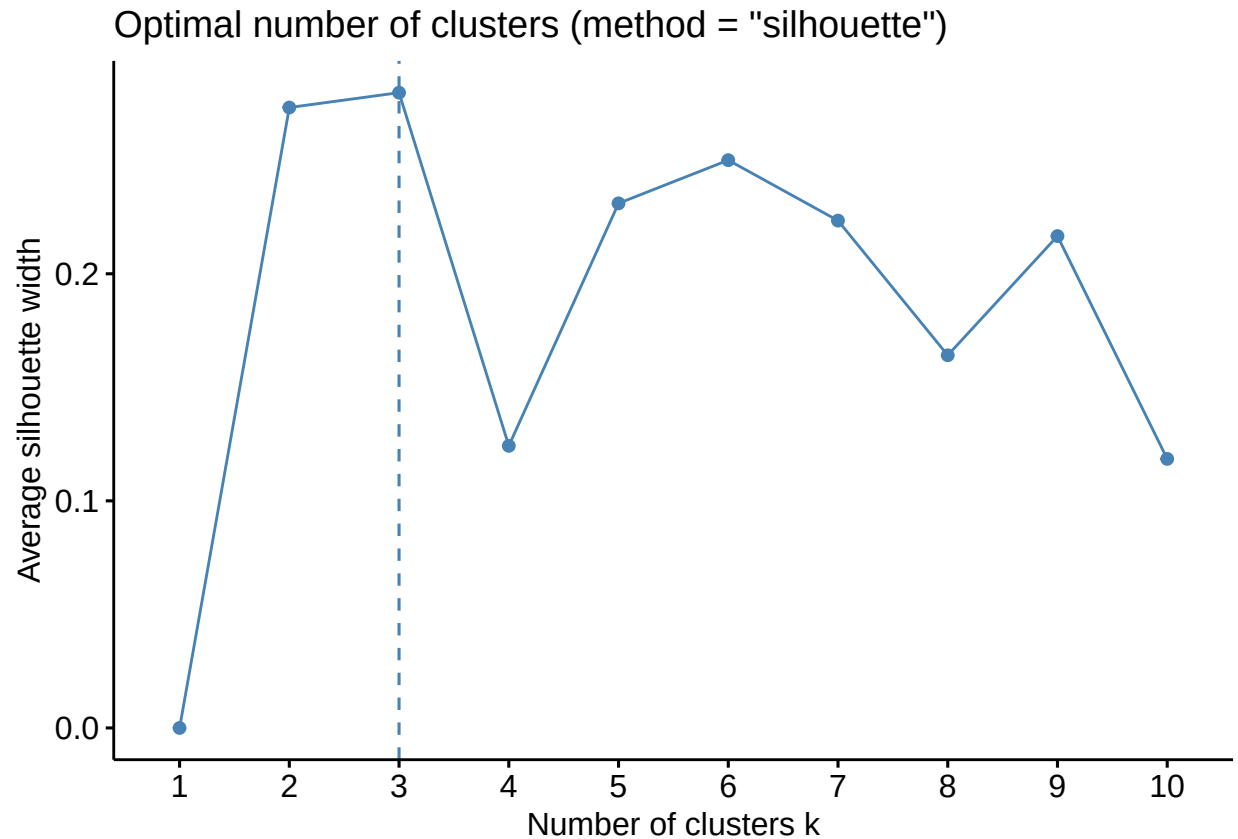
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```
data <- read.csv("C:/Users/brumb/OneDrive/Documents/R/Pharmaceuticals.csv")  
numeric_data <- data[, 3:11]  
scaled_data <- scale(numeric_data)
```

```
fviz_nbclust(scaled_data, kmeans, method = "wss")
```



```
fviz_nbclust(scaled_data, kmeans, method = "silhouette")
```



To determine the appropriate number of clusters, both the elbow method and the silhouette method were used. The elbow plot showed a clear bend at $k = 3$, indicating diminishing returns in reducing within-cluster variation beyond this point. The silhouette method also supported $k = 3$, suggesting that this number provides well-separated and cohesive clusters.

The k-means algorithm was then applied with:

$k = 3$ clusters $nstart = 25$ to improve stability and avoid poor local solutions due to random initialization

All variables were weighted equally, as no prior justification existed to emphasize one financial metric over another.

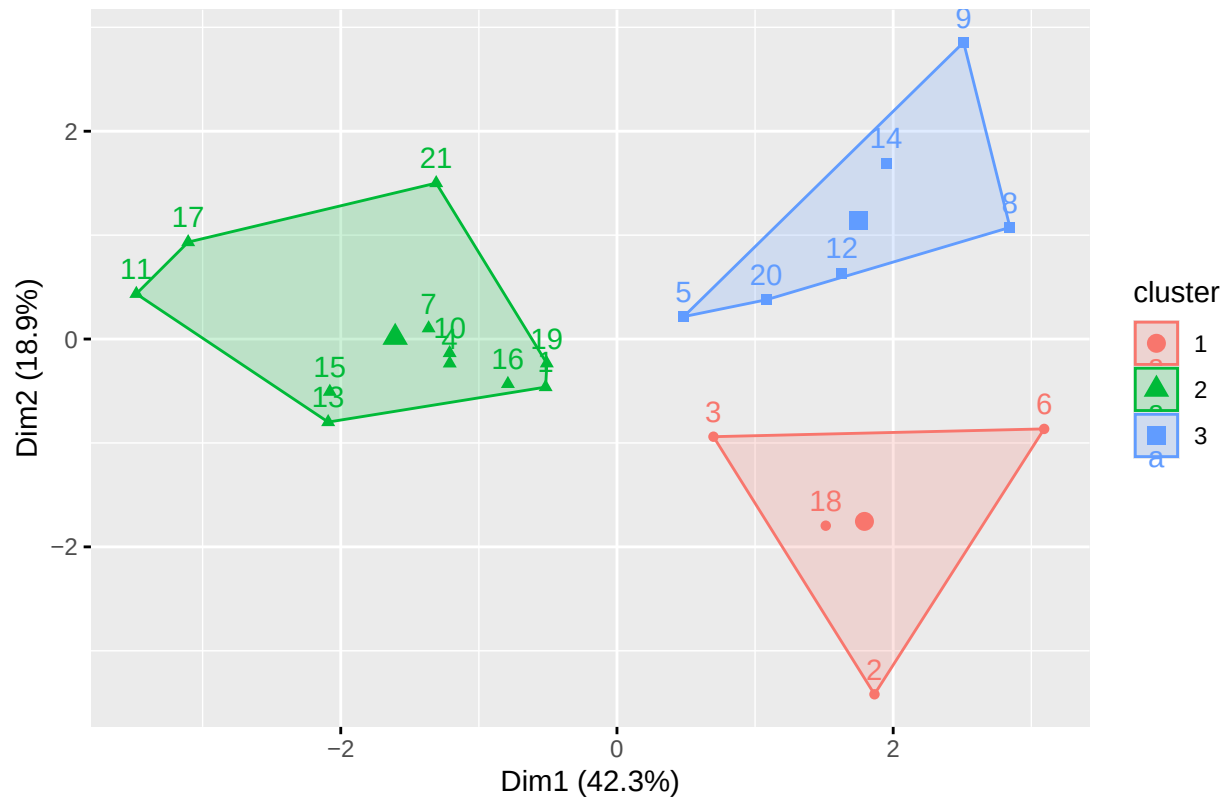
```
set.seed(123)

kmeans_result <- kmeans(scaled_data, centers = 3, nstart = 25)

data$Cluster <- kmeans_result$cluster

fviz_cluster(kmeans_result, data = scaled_data)
```

Cluster plot



```
aggregate(numeric_data, by = list(Cluster = data$Cluster), mean)
```

```
##   Cluster Market_Cap      Beta PE_Ratio  ROE      ROA Asset_Turnover  Leverage
## 1      1    21.75500 0.5950000 46.90000 11.3  5.100000      0.7500000 0.3050000
## 2      2    97.11364 0.4336364 20.95455 35.7 14.954545      0.8000000 0.3254545
## 3      3     9.23500 0.6483333 19.43333 17.3  5.983333      0.4833333 1.2500000
##   Rev_Growth Net_Profit_Margin
## 1    7.01000      6.65000
## 2   10.16455     20.17273
## 3   23.49000     13.51667
```

Cluster 1 Medium market cap High P/E ratio Low ROE and ROA Low revenue growth Low profit margins
These firms have weaker performance and may be overvalued.

Cluster 2 Largest market cap High ROE and ROA High profit margins Moderate growth
These are large, stable, and highly profitable firms.

Cluster 3 Small market cap Highest revenue growth Higher leverage Moderate profitability
These firms are growing quickly but are riskier.

```
table(data$Cluster, data$Location)
```

```
##
##   CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
```

```
##      1      1      0      1      0      0      1      1
##      2      0      0      0      0      0      1      2      8
##      3      0      1      0      1      0      0      0      4
```

```
table(data$Cluster, data$Exchange)
```

```
##
##      AMEX  NASDAQ  NYSE
##      1      0      0      4
##      2      0      0     11
##      3      1      1      4
```

```
table(data$Cluster, data$Median_Recommendation)
```

```
##
##      Hold Moderate Buy Moderate Sell Strong Buy
##      1      2      1      0      1
##      2      6      3      2      0
##      3      1      3      2      0
```

Location Cluster 2 is mostly US and UK Cluster 3 includes more international firms (France, Ireland) Cluster 1 is mixed but smaller

Stock Exchange Cluster 2 firms are all on the NYSE Cluster 3 includes AMEX and NASDAQ Cluster 1 firms are also on NYSE

Median Recommendation Cluster 2 has mostly Hold and Moderate Buy Cluster 3 has more mixed ratings, including Moderate Sell Cluster 1 includes a Strong Buy but also weaker ratings

Cluster 1

Underperforming / Overvalued Firms

Cluster 2

Large Established Pharma Leaders

Cluster 3

High-Growth, High-Risk Firms

Three main groups have been identified in the industry. There are, large, profitable firms that are stable and dominate the market, high-growth firms that are smaller but riskier, and lower-performing firms with weaker financial results. Overall, the clusters show clear differences in size, profitability, growth, and risk, helping better understand the structure of the industry.